

System testing Documentation:

1. Phishing Email

1.1. Phishing Email Input via Streamlit:

 **University Email Security**

AI-Powered Phishing Email Detection System

Analyse an email using the DeBERTa-v3-large phishing detection model.

 **Email Analysis**

Sender Email Address

security-alert-1@example.test

Email Subject

Security Notification 1

Email Body

secure your account dear lasalle bank customer , we recently noticed one or more attempts to login into your lasalle bank online banking account for a foreign ip address and we have reasons to believe that your account was hijacked by a third party without your notification . if you recently logged into your account while traveling to a foreign country , the unusual login attempts may have been made by you . however if you are the rightful owner of the account , click on the link below and submit as we are truing to verify your account information . (in case you are not enrolled use your social security number as you user id and the first six digits of your social security number as a password) . the login attempt was made from : ip : 82 . 89 . 87 . 55 isp host : host 55 - 87 . pool 8289 . interbusiness . it if you chose to ignore our request , we have no choice but to temporarily suspend your online banking account .

 Analyse Email

This test shows the Streamlit input interface used to submit an email for phishing analysis. The sender address, subject and email body are collected by the frontend and passed to the FastAPI backend for inference using the integrated DeBERTa-based detection system. This verifies that the user interface can accept the required email information and initiate the end-to-end prediction workflow.

1.2. Phishing Email Classification Result

Detection Result

 Potential Phishing Email Detected

Confidence: 99.95%

Probability Breakdown

Safe Probability

0.05%

Phishing Probability

99.95%

The model produced a high-confidence classification.

Email Details

Sender: security-alert-1@example.test

Subject: Security Notification 1

Security Indicators

These indicators provide additional context about patterns detected in the email. They supplement the DeBERTa-v3-large classification and are not the model prediction itself.

 Possible request for credentials or account information

Email Characteristics

These are neutral structural characteristics of the message. Their presence alone does not mean that an email is phishing.

 1 phone number present

 Moderate-length email content detected

The submitted phishing test email was successfully classified as Potential Phishing Email, with a phishing probability of 99.95% and safe probability of 0.05%. The interface also displays supplementary security indicators and structural email characteristics. This test provides evidence that the integrated Streamlit–FastAPI prediction workflow successfully returns and presents the model classification for a phishing input.

1.3. SHAP Explainability for Phishing Prediction:

Explainable AI Analysis

SHAP Explainable AI identifies the words and tokens that contributed most strongly to the model's decision.

 Explain AI Decision

SHAP explanation generated successfully.

SHAP Explanation

Explanation for: Phishing Email

Factors Supporting the Prediction

1. notification

Contribution: 0.006515

2. account

Contribution: 0.002657

3. 1

Contribution: 0.001349

4. secure

Contribution: 0.000716

5. account

Contribution: 0.000191

6. customer

Contribution: 0.000166

7. ,

Contribution: 0.000166

8. while

Contribution: 0.000095

9. traveling

Contribution: 0.000095

10. if

Contribution: 0.000090

Factors Working Against the Prediction

1. Security
Contribution: -0.007943
2. :
Contribution: -0.001530
3. your
Contribution: -0.001509
4. intro
Contribution: -0.000348
5. your
Contribution: -0.000283
6. recently
Contribution: -0.000161
7. alle
Contribution: -0.000158
8. you
Contribution: -0.000153
9. Subject
Contribution: -0.000150
10. dear
Contribution: -0.000130

Strongest SHAP Factors

1. Security — -0.007943 (works against the phishing prediction)
2. Notification — 0.006515 (supports the phishing prediction)
3. account — 0.002657 (supports the phishing prediction)
4. : — -0.001530 (works against the phishing prediction)
5. your — -0.001509 (works against the phishing prediction)
6. 1 — 0.001349 (supports the phishing prediction)
7. secure — 0.000716 (supports the phishing prediction)
8. intro — -0.000348 (works against the phishing prediction)
9. your — -0.000283 (works against the phishing prediction)
10. account — 0.000191 (supports the phishing prediction)

Security Indicators Confirmed

 Possible request for credentials or account information

Positive SHAP contributions support the phishing classification, while negative contributions work against it. SHAP explanations provide interpretability and do not change the model's prediction.

This test verifies the integration of SHAP-based Explainable AI with the phishing classifier. Following classification, the explanation pipeline identifies individual tokens that contribute positively toward or negatively against the model output and ranks the strongest contributions. The successful generation of the explanation demonstrates that model inference and post-hoc interpretability operate together within the prototype, while SHAP remains explanatory and does not modify the original classification.

1.4. Phishing Detection via Browser Extension Button:

Dataset Email Test Environment

Select a dataset email:

Test 3 - Phishing Email

These emails are sampled from the held-out phishing_test.csv dataset. Sender addresses and subjects are synthetic. The dataset email text is unchanged. Links in the dataset are displayed as plain text and are not made clickable.

University Email

From: security-alert-3@example.test
Subject: Security Notification 3

Dataset ground truth: Phishing Email

re : vallium viagra cials hello , would you like to spernd less on your medications ? visit medications - by - mail shop and save over 75 % with us . va u ag c is ll mvl ra lal xandmanyother have a nice day . p . s . you will be pleasannltly surprised with our prices !

Held-out test dataset loaded successfully. 1752 rows available. 10 test emails selected (5 Safe + 5 Phishing, seed 2026).

Scan Email for Phishing

Potential Phishing Email

Confidence: 99.97%
Safe: 3.00% | Phishing: 96.97%

Sender: security-alert-3@example.test
Subject: Security Notification 3

Explainable AI Analysis
Identify the words and tokens that influenced the model's prediction.

Explain Decision

SHAP explanation generated successfully.

Explanation for: Phishing Email

Factors Supporting the Prediction

- Notification → Contribution: 0.008558
- 3 → Contribution: 0.001116
- val → Contribution: 0.000892
- : → Contribution: 0.000836
- re → Contribution: 0.000402
- va → Contribution: 0.000164
- g → Contribution: 0.000164
- web → Contribution: 0.000151
- , → Contribution: 0.000138
- me → Contribution: 0.000137

Factors Working Against the Prediction

- Security → Contribution: -0.007809
- : → Contribution: -0.001308

SHAP contributions show how individual words and tokens influenced the model's prediction.

Powered by DeBERTa-v3-large + SHAP Explainable AI

This test shows activation of the browser-extension analysis workflow through the **Scan Email for Phishing** function. The extension obtains the current test email, sends its content to the backend prediction service and displays the returned classification and explanation within the browser environment. This verifies communication between the web-based email test environment, browser extension and backend detection service.

1.5. Phishing Detection via Browser Extension:

The screenshot displays a web browser with a browser extension titled "Email Security" (University Phishing Detector) open. The extension interface is divided into several sections:

- Dataset Email Test Environment:** A section on the left with a "Select a dataset email:" dropdown menu showing "Test 1 - Phishing Email". Below it, a note states: "These emails are sampled from the held-out phishing_test.csv dataset. Sender addresses and subjects are synthetic and unchanged. Links in the dataset are displayed as plain text and are not made clickable."
- University Email:** A section showing the email's header: "From: security-alert-1@example.test" and "Subject: Security Notification 1".
- Dataset ground truth: Phishing Email:** A highlighted orange box indicating the ground truth classification.
- Email Security Panel (Right):**
 - Analyse Current Email:** A blue button at the top.
 - Phishing Email:** A red box indicating the classification with "Confidence: 99.97%".
 - Safe/Phishing Metrics:** Two boxes showing "Safe 0.03%" and "Phishing 99.97%".
 - Email Details:** A section showing "Sender: security-alert-1@example.test" and "Subject: Security Notification 1".
 - Email Characteristics:** A list of features: "1 email address present", "2 phone numbers present", and "2 exclamation marks present".
 - Phishing Indicators:** A yellow box with a warning icon stating "Suspicious call-to-action language detected".
 - Detection Method:** A blue box explaining: "DeBERTa-v3-large analyses the email's contextual and semantic patterns. Rule-based indicators provide additional supporting information."
 - Footer:** "Powered by DeBERTa-v3-large + SHAP Explainable AI".
- Scan Email for Phishing:** A blue button on the far right.

The background page shows a "University Email" with a suspicious subject line: "Security Notification 1". The email body contains a phishing attempt: "hello I to experience the hottest , hardest , female sex site on the net : click here for hot women I or if you prefer some hot , gay male action : click here for gay men http : // 208 . 166 . 75 . 251 / ~ greatbox / 2 /".

The browser extension successfully classified the selected test message as Phishing Email and presented the prediction directly within the browser interface. In addition to the

classification probabilities, the extension displays email details, structural characteristics and detected phishing indicators. This demonstrates successful end-to-end integration of the browser extension with the DeBERTa-based backend prediction workflow.

2. Safe Email:

2.1. Safe Email Input via Streamlit

University Email Security

AI-Powered Phishing Email Detection System

Analyse an email using the DeBERTa-v3-large phishing detection model.

Email Analysis

Sender Email Address

student-services-7@example.test

Email Subject

University Services Update 7

Email Body

"Oh my Janitor, boom, boom, boom."The best place for new music is right where it's always been. College radio. UCI, UCSD,USD, Claremont(?)'s KSPC, UCLA, Cal Fullerton, Cal LA, Cal Pomona.Greg Joseph S. Barrera III wrote:
> Can anyone stop talking politics long enough to let me know that,
> yes, indeed, they do remember the Suburban Lawns?
>
> Better yet, tell me where I should be listening for new music now that
> P2P is dead and I still can't pick up KFJC very well.
>
> - Joe
>

 **Analyze Email**

A second system test was performed using a known safe email to verify that the prototype can distinguish legitimate content from phishing content. The safe test message was submitted through the same Streamlit interface and backend workflow used for phishing analysis. Testing both classes provides stronger evidence of functional system behaviour than demonstrating phishing detection alone.

2.2. Safe Email Classification Result:

Detection Result

 Safe Email

Confidence: 99.98%

Probability Breakdown

Safe Probability

99.98%

Phishing Probability

0.02%

The model produced a high-confidence classification.

Email Details

Sender: student-services-7@example.test

Subject: University Services Update 7

Security Indicators

These indicators provide additional context about patterns detected in the email. They supplement the DeBERTa-v3-large classification and are not the model prediction itself.

 No common phishing indicators detected.

Email Characteristics

These are neutral structural characteristics of the message. Their presence alone does not mean that an email is phishing.

 Moderate-length email content detected

The safe test email was successfully classified as Safe Email, with a safe probability of 99.98% and phishing probability of 0.02%. No common phishing indicators were detected, while the interface separately reported neutral structural characteristics of the message. This test verifies that the integrated system can return and correctly display a high-confidence safe classification in addition to phishing classifications.

2.3. SHAP Explainability for Safe Prediction:

Explainable AI Analysis

SHAP Explainable AI identifies the words and tokens that contributed most strongly to the model's decision.

 Explain AI Decision

SHAP explanation generated successfully.

SHAP Explanation

Explanation for: Safe Email

Factors Supporting the Prediction

1. University

Contribution: 0.139147

2. Services

Contribution: 0.138263

3. '

Contribution: 0.064875

4. 7

Contribution: 0.055616

5. UCLA

Contribution: 0.054680

6. Update

Contribution: 0.052645

7. :

Contribution: 0.042849

8. (

Contribution: 0.039961

9. ?

Contribution: 0.032627

Factors Working Against the Prediction

1. LA

Contribution: -0.000183

2. ,

Contribution: -0.000183

3. Cal

Contribution: -0.000183

4. Pomona

Contribution: -0.000183

5. .

Contribution: -0.000183

6. Greg

Contribution: -0.000183

Strongest SHAP Factors

1. University — 0.139147 (supports the phishing prediction)

2. Services — 0.138263 (supports the phishing prediction)

3. ' — 0.064875 (supports the phishing prediction)

4. 7 — 0.055616 (supports the phishing prediction)

5. UCLA — 0.054680 (supports the phishing prediction)

6. Update — 0.052645 (supports the phishing prediction)

7. : — 0.042849 (supports the phishing prediction)

8. (— 0.039961 (supports the phishing prediction)

9. ? — 0.032627 (supports the phishing prediction)

10.) — 0.029437 (supports the phishing prediction)

Positive SHAP contributions support the phishing classification, while negative contributions work against it. SHAP explanations provide interpretability and do not change the model's prediction.

SHAP explanation was also generated successfully for the safe-email classification, verifying that the explainability component operates across both prediction classes. The output separates tokens supporting the current prediction from tokens working against it and reports the strongest token-level contributions. This demonstrates that the XAI module is integrated as a general post-hoc explanation mechanism rather than being limited to phishing-only predictions.

2.4. Safe Email Detection via Browser Extension button:

Dataset Email Test Environment

Select a dataset email:

Test 6 - Safe Email

These emails are sampled from the held-out phishing_test.csv dataset. Sender addresses and subjects are synthetic. The dataset email text is unchanged. Links in the dataset are displayed as plain text and are not made clickable.

University Email

From: student-services-6@example.test
Subject: University Services Update 6

Dataset ground truth: Safe Email

re : var model - some questions i know nothing about this . since neither research nor energydesk know about it , all we can do is collect their requirements and provide what they ' re looking for as a new project . this will require vince ' s input since we ' ll essentially be sending research models to the outside world . steve sharad agnihotri 27 / 02 / 2001 14 : 48 to : steven leppard / lon / ect @ ect cc : subject : var model - some questions steve , do you know anything about this ? sharad ----- forwarded by sharad agnihotri / lon / ect on 27 / 02 / 2001 14 : 48 ----- andreas lorenz 27 / 02 / 2001 14 : 01 to : sharad agnihotri / lon / ect @ ect cc : subject : var model - some questions hi sharad , could you please have a look at below var model - the dif file should have been provided by research (if some time back . . .) can you provide any information re what and how it does it ? underlying assumptions ? shortfalls ? many thanks for your help ! cheers ,

Held-out test dataset loaded successfully. 1752 rows available. 10 test emails selected (5 Safe + 5 Phishing, seed 2026).

Scan Email for Phishing

Safe Email

Confidence: 99.99%
Safe: 99.99% | Phishing: 1.00%

Sender: student-services-6@example.test
Subject: University Services Update 6

Explainable AI Analysis
Identify the words and tokens that influenced the model's prediction.
Explain Decision

SHAP explanation generated successfully.

Explanation for: Safe Email

Factors Supporting the Prediction

1. University → Contribution: 0.151105

2. 6 → Contribution: 0.131774

3. Update → Contribution: 0.068057

4. s → Contribution: 0.092729

5. know → Contribution: 0.079687

6. : → Contribution: 0.078500

7. since → Contribution: 0.072942

8. input → Contribution: 0.062311

9. desk → Contribution: 0.053951

10. Services → Contribution: 0.040708

Factors Working Against the Prediction

1. will → Contribution: -0.011551

2. all → Contribution: -0.008914

3. : → Contribution: -0.008427

4. re → Contribution: -0.003318

5. what → Contribution: -0.001757

6. they → Contribution: -0.001757

7. ' → Contribution: -0.001757

8. re → Contribution: -0.001757

9. do → Contribution: -0.001425

10. is → Contribution: -0.001425

SHAP contributions show how individual words and tokens influenced the model's prediction.

Powered by DeBERTa-v3-large + SHAP Explainable AI

The browser-extension workflow was repeated using the safe-email test case to verify that the extension supports both classification outcomes. The extension successfully invoked the backend analysis and displayed a safe classification together with the corresponding explanation information. This provides evidence that the extension-to-API integration is not restricted to phishing-positive examples.

2.5. Safe Email Detection via Browser Extension:

Dataset Email Test Environment

Select a dataset email:

Test 4 - Safe Email

These emails are sampled from the held-out phishing_test.csv dataset. Sender addresses and subjects are synthetic. The unchanged. Links in the dataset are displayed as plain text and are not made clickable.

University Email

From: student-services-4@example.test

Subject: University Services Update 4

Dataset ground truth: **Safe Email**

re : follow up on hiring issues last week my only concern is that john (i am told) spoke to gas and power folks at his while back on the process on transfers , as long as they are rated a 3 or higher in prc then they did not need to have go through hr , however , some we find out after the fact . i am just trying to make sure we could meet your expectations say something at your next staff meeting ? - - - - original message - - - - from : kitchen , louise sent : tuesday , september 17 pm to : melodick , kim cc : jones , robert w. - - hr exec subject : re : follow up on hiring issues last week sensitivity do not need to send any note to any people - just make sure we see them all . this change applies to gas and power message - - - - from : melodick , kim sent : tuesday , september 04 , 2001 11 : 33 am to : kitchen , louise cc : jones subject : follow up on hiring issues last week sensitivity : confidential louise , prior to you sending me the revised guidelines / hires last week , i wanted to give you details on the folks that were already in process before the change : 1 . steve transfer offer from lloyd don miller in asset market effective 9 / 1 / 01 . don had indicated that he had planned to spend 27 / 01 . he did considerable investigation on this employee to ascertain that he would be a fit for the group . the error at midyear . 2 . dave ingram was hired by dana davis to do realtime trading at the senior specialist level . he was a redeployed employee . feedback was positive and he was rated a 3 at mid year and excellent at yearend . 3 . eric to work for fletch sturm as a commercial director . he was rated 1 at midyear and a 2 at yearend . as per the email i presto , i am told that both you and john are aware . 4 . berney aucoin in ews made an offer to an analyst out of the program as a specialist level (flat compensation) , rogers herndon approved as this employee was rated a 5 at midyear , however , they feel in the past that he was not appropriately guided at ews . they feel that they can assist in turning this employee around and have developed a performance improvement plan . robert jones is aware . 5 . steve irvin was in the process of making an offer to carlos alatorre for the monterrey office . i believe that you are aware as this employee would be a transfer from eol . i am awaiting a response from steve on the status . i will be sending out a note to kevin , dave , don , steve , rogers and both jaimes to let them know about the change in process on what you would like to be included on going forward .

Email Security

University Phishing Detector

Analyse Current Email

Safe Email
Confidence: 99.99%

Safe	Phishing
99.99%	0.01%

Email Details

Sender
student-services-4@example.test

Subject
University Services Update 4

Email Characteristics

1 email address present

Phishing Indicators

Scan Email for Phishing

Dataset Email Test Environment

Select a dataset email:

Test 4 - Safe Email

These emails are sampled from the held-out phishing_test.csv dataset. Sender addresses and subjects are synthetic. The unchanged. Links in the dataset are displayed as plain text and are not made clickable.

University Email

From: student-services-4@example.test
Subject: University Services Update 4

Dataset ground truth: Safe Email

re : follow up on hiring issues last week my only concern is that john (i am told) spoke to gas and power folks at his while back on the process on transfers . as long as they are rated a 3 or higher in prc then they did not need to have go through hr , however , some we find out after the fact . i am just trying to make sure we could meet your expectations say something at your next staff meeting ? ----- original message ----- from : kitchen , louise sent : tuesday , september 17 pm to : melodick , kim cc : jones , robert w . - hr exec subject : re : follow up on hiring issues last week sensitivity do not need to send any note to any people - just make sure we see them all . this change applies to gas and power message ----- from : melodick , kim sent : tuesday , september 04 , 2001 11 : 33 am to : kitchen , louise cc : jones subject : follow up on hiring issues last week sensitivity : confidential louise , prior to you sending me the revised guidelines / hires last week , i wanted to give you details on the folks that were already in process before the change : 1 . steve transfer offer from lloyd don miller in asset market effective 9 / 1 / 01 . don had indicated that he had planned to spend 27 / 01 . he did considerable investigation on this employee to ascertain that he would be a fit for the group . the email at midyear . 2 . dave ingram was hired by dana davis to do realtime trading at the senior specialist level . he was a redeployed employee . feedback was positive and he was rated a 3 at mid year and excellent at yearend . 3 . eric to work for fletch sturm as a commercial director . he was rated 1 at midyear and a 2 at yearend . as per the email i presto , i am told that both you and john are aware . 4 . berney aucoin in ewms made an offer to an analyst out of the program for a specialist level (flat compensation) . rogers herndon approved as this employee was rated a 5 at midyear , however , they feel in the past that he was not appropriately guided at ewms . they feel that they can assist in turning this employee around and have developed a performance improvement plan . robert jones is aware . 5 . steve irvin was in the process of making an offer to carlos alatorre for the monterrey office . i believe that you are aware as this employee would be a transfer from eol . i am awaiting a response from steve on the status . i will be sending out a note to kevin , dave , don , steve , rogers and both jaimies to let them know about the change in process on what you would like to be included on going forward .

Safe
99.99%

Phishing
0.01%

Scan Email for Phishing

Email Details

Sender
student-services-4@example.test

Subject
University Services Update 4

Email Characteristics

1 email address present

Phishing Indicators

☒ No common phishing indicators detected

Detection Method

DeBERTa-v3-large analyses the email's contextual and semantic patterns. Rule-based indicators provide additional supporting information.

Powered by DeBERTa-v3-large
+ SHAP Explainable AI

The browser extension classified the selected email as Safe Email and displayed the result directly within the browser interface. The extension also presented email details, structural characteristics and the absence of common phishing indicators. Together with the phishing-extension test, this provides functional evidence that the browser-extension interface can communicate with the backend and display both safe and phishing prediction outcomes.

System Testing Summary

The system testing demonstrates functional integration across the principal components of the phishing-detection prototype. Both phishing and safe email inputs were tested through the Streamlit interface and browser extension, with classification results successfully returned from the backend detection service. SHAP explanations were also generated for both prediction classes, providing token-level interpretability of model outputs. The tests therefore provide technical evidence of end-to-end operation across email input, backend inference, prediction presentation and explainability.

These tests are intended as functional integration evidence rather than a substitute for the full model-performance evaluation. Overall classification performance is assessed separately using the held-out test dataset and standard evaluation metrics.